TANMAY HINGE

Available for doctoral candidacy

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RESEARCH INTERESTS

Learning analytics, and machine learning applied to the sciences.

EDUCATION

M.S. Computer Science, California State University, Chico, United States 2022 to May 2024

- Coursework in Applied Machine Learning, Applied Computer Vision, Applied Graph Theory, Algorithms and Computability, Research Methods in Computer Science, and Advanced Computer Security.
- Capstone, Statistical Methods for Assessing Candidate Employability.

B.E. Computer Engineering, Pillai College of Engineering, University of Mumbai, India 2018 to June 2022

- Coursework in Machine Learning, Natural Language Processing, Digital Signal and Image Processing, Data Warehousing and Mining, Big Data and Analytics, Distributed Computing, and Applied Mathematics I to IV.

RESEARCH

Predicting treatment delays of more than 90 days in prostate cancer (preprint and code) 2026

- Cohort of 330,827 men treated with surgery or radiotherapy in the US SEER 17 registries, 2010 to 2022. The question is how much county rurality, county income and marital status add to clinical information in predicting a wait of more than 90 days. The share waiting that long rose from 36.0% in 2010 to 52.3% in 2022.
- Penalised logistic regression and LightGBM, cross-fitted in ordered feature blocks within risk groups, with bootstrap intervals. Registry data predicted delay weakly, AUC 0.592 to 0.663 where 0.5 is chance. The area and marital block added 0.62 percentage points of log-loss skill (95% interval 0.39 to 0.87) with logistic regression and 0.99 (0.73 to 1.30) with LightGBM, higher being better.
- Public protocol with an amendment log, small-count suppression under the SEER data use agreement, and a build step that flags any number in the paper absent from the analysis reports. 250 tests.

Probing a frozen rPPG encoder for identity under input degradation (report and code) 2026

- Froze two released camera-based heart-rate models and probed their representation on 122 synthetic videos with known pulse. Identity of 120 subjects held at 100% top-1 accuracy against 0.83% for guessing, down to a 4 by 4 block average of the skin, and fell to 12.5% only once the face-shaped mask was deleted. Blurring a face does not anonymise it.
- Heart-rate error of 25.3 beats per minute against 28.8 for a constant predictor, lower being better, was not separable from it once bootstrapped. Preprocessing checked bit-identical against rPPG-Toolbox. 107 tests.

Benchmarking Graph Neural Network Attribution on Synthetic Tissue Graphs (report and code) 2026

- Synthetic tumour tissue where the cells driving the label are known. GNNExplainer recovers them at 0.037 to 0.049 average precision against a random baseline of 0.058, higher being better. Integrated gradients alone survives weight randomisation, at 0.205 against 0.169 for a heuristic that consults no model.
- GNNExplainer's randomisation rank correlation moves from 0.89 to 0.29 on optimisation budget alone, so the control at the common default is unconverged. Two of my own verdicts were retracted and re-measured, and both corrections stay in the repository. 92 tests across five gated phases.

Learning Curves for Kernel Ridge Regression on Molecular Excitonic Couplings (report and code) 2026

- Across 40,000 molecular pairs, new pairs of known molecules reach 0.008 mean absolute error while unseen molecules are roughly 600 times worse, a gap only grouped cross-validation exposes. Gaussian process posterior variance separates the two cases with no labels.

Probing Masked-Sequence Pretraining on Synthetic Mixed Spectra (report and code) 2026

- A 544,000-parameter transformer pretrained on 20,000 simulated spectra. Fine-tuning and frozen probing disagree. Frozen probes decode concentration at 0.150 mean absolute error against 0.239 for random initialisation, lower being better. 46 seeded tests.

Benchmarking an AI SOC Analyst on Synthetic Intrusion Telemetry (report and code) 2026

- An LLM analyst scored on synthetic security logs by two graders that are never averaged. It reported an intrusion in 23 of 24 clean windows, and a random citer scored 1.000 on grounding against its 0.639, higher being better.

Earlier robotics and perception work 2024 to 2025

- SAC reached 78% peg-in-hole success against 61% for PPO in MuJoCo and Isaac Gym, holding 65% under domain randomisation. Visual-inertial SLAM with corrected ATE of 0.15 m over a 450 m trajectory at 24 Hz on a Jetson Orin Nano.

PROFESSIONAL EXPERIENCE

Computer Science Faculty, Inver Hills Community College, Minnesota, United States Jul 2024 to Jul 2026

- Full time. Designed, built and taught the entire Applied AI certificate programme, and taught four core courses in introductory computing, C++, Python and Java.
- Mentored more than 1,000 students individually, and ran outreach to five local high schools on computer science careers. Taught K-12 robotics in Minnesota alongside the faculty role.

AI Researcher, Regace AI Feb 2026 to present

- Founded Regace AI and built its retrieval-augmented generation engine over regulatory text, with hybrid dense and keyword retrieval, reranking and citation-grounded generation.
- Designed the evaluation harness (curated question set, retrieval precision and recall, grounding checks) that decides whether a pipeline change is real. In production over more than 10,000 indexed documents with two pilot clients.

Co-Founder and Technical Lead, Merit Labs May 2026 to present

- Built MeritQ, the company's exam delivery platform, from scratch on Next.js, PostgreSQL and Docker with containerised integration tests.
- Built a human-in-the-loop LLM pipeline that turns student responses into individual reports on strengths, gaps and career direction for Classes 5 to 12, iterated against expert graders.

Tutor and Laboratory Assistant, California State University, Chico Aug 2023 to May 2024

- Laboratory assistant for Programming and Algorithms II and for Data Structures and Algorithms. Tutored C++, algorithms and computer security.

Software Engineer Intern, VettX Inc., California May 2023 to Aug 2023

- Built a client-onboarding platform in React and a Chrome extension that shortened seller outreach by roughly 40 percent.

Instructor and Curriculum Developer, Pillai College of Engineering and CampK12 2019 to 2022

- Workshops in C++, competitive programming and Unity game development, and programming curricula for school-age cohorts.

TECHNICAL SKILLS

- **Machine learning.** PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, LightGBM, transformers, self-supervised pretraining, linear probing, attribution under randomisation controls, Gaussian processes, knowledge graphs, vision-language models, LLM fine-tuning (SFT, LoRA), RAG and retrieval evaluation.
- **Statistics.** Penalised logistic and ridge regression, cross-fitting, grouped cross-validation, bootstrap intervals, calibration, standardisation, permutation and label-shuffle controls, survey instrument design.
- **Engineering.** Python at production grade, C++, Java, Rust, TypeScript, Swift, C#; NumPy, SciPy, pytest; Next.js, PostgreSQL, MongoDB; Docker, Kubernetes, CI/CD, Git; AWS, GCP, Azure; ROS 2, OpenCV, Unity with AR Foundation and OpenXR; SEER*Stat; LaTeX.
- **Languages.** English (native, fluent), Marathi (native, fluent), Swedish (beginner), German (beginner).

ACADEMIC REFEREES

Samuel B. Siewert, Professor of Computer Science, California State University, Chico
sbsiewert@csuchico.edu · Graduate course instructor, Applied Computer Vision

Richard C. Tillquist, Assistant Professor of Computer Science, California State University, Chico
rctillquist@csuchico.edu · Graduate course instructor, Applied Graph Theory

Tyson R. Henry, Professor of Computer Science (retired), California State University, Chico
TRHenry@csuchico.edu · Department Chair, and supervisor for the tutoring and assistant appointments